

Bruno Guzzo

System Engineer — AI & ML Specialist

Work Experience

System Engineer

April 2026 – Present

AWS (EUSC, Managed Operations), Dublin

- Providing system engineering and operational support within the **European Union Sovereignty Cloud (EUSC)** Managed Operations, ensuring high availability and compliance for critical infrastructure. Responsible for managing and scaling **Amazon DynamoDB** and **Amazon Redshift** clusters.

AI Specialist (Mid-Senior Consultant)

January 2025 – April 2026

Go Reply, Turin

- Coordinated a multi-company effort to gather customer requirements for a **research-oriented healthcare platform**, enabling centralized and rapid clinical or genomic **data retrieval**. Lead project key phases from defining a complete **functional analysis** document, design a **cloud-native architecture** and implementation through sprint planning.
- Directed a cross-functional team of developers in delivering a **B2C cloud-native mobile application** enabling after-sales support, digital documentation access via **Gen-AI RAG chat-bot** and seamless multi-cloud **CRM integration**. Guided the adoption of a distributed serverless **micro-services** architecture coordinated using **Google Pub/Sub** to enforce **fault isolation** and scalability.
- Designed and implemented a scalable **legal treaties search engine** and **RAG solution** using **Vertex AI** and **BigQuery** with a highly efficient and concurrent data ingestion pipeline, enabling insight generation and optimization of customer workflow. Successfully improved data processing speed by **94%** after adopting a serverless cloud-native master/worker architecture relying on **Cloud Task** and **Cloud Run** for concurrent workloads distribution.
- Advised clients on **AI and agent-based system** integration, facilitating the adoption of Google Cloud enterprise ecosystem via **custom-tailored RFP solutions** to improve critical business processes. Increased client adoption rate of Vertex AI by **30%**.
- Managed **strategic client engagement** within the global development industry, coordinating **multi-project cloud advisory** and **resource allocation** that mobilized **20%** of the BU workforce and drove **30%** of unit profitability.
- Mentored junior engineers, **promoted agile best practices** through knowledge-sharing events, and **coordinated sprint reviews** and retrospectives to accelerate team performance and continuous improvement. Successfully decreased code-review time by **60%**.

Cloud Engineer (Consultant)

June 2022 – December 2024

Go Reply, Turin

- Contributed to the development of a digital platform for the **agriculture and food sector**, enhancing **data collection and analytics** capabilities to support evidence-based decision-making on sustainable farming and antimicrobial resistance. Lead the platform **migration to Google Cloud** relying on IAC best-practices as full **Terraform foundation** and **GCP Fabric** modules to ensure custom service configuration and resources tracking. Successfully reduced infrastructural fixed-costs by **30%**.
- Designed efficient multi-stage **Git-Hub** and **Git-Lab CI/CD pipeline** for quickest application deployment to production using **Cloud Build** for **Docker** image building and a fine-grained configured **Cloud Run** instance (or Firebase hosting) for computing. Bug-free releases are accomplished leveraging a comprehensive test-cases suite targeting both back-end, using **Junit** or **PyTest**, and front-end, using **Selenium** to mimic end-user actions.
- Modernized a large-scale **telecommunications cybersecurity platform**, improving scalability and enhancing user experience through **software architectural refactoring**, cloud integration, and automation to strengthen **service reliability**.

Personal & Academic Projects

MSc Thesis: Anomaly Detection with Deep Autoencoders

- Developed a novel Autoencoder (AE-SAD) for semi-supervised anomaly detection, significantly improving accuracy (AUC) and reducing training time by addressing class imbalance.

Transformer-GNN for Link Prediction

- Engineered a scalable data pipeline for large-scale graph dataset generation from Wikipedia. Trained and assessed Graph Attention Network (GAT) models for link prediction.

Education

MSc, AI & ML Engineering

Oct 2022 – July 2025

UNICAL, Rende (CS), Italy

BSc, Computer Engineering

Oct 2018 – June 2022

UNICAL, Rende (CS), Italy



13 Jan 2000

+393398305458

brunoguzzo18@gmail.com

github.com/bGuzzo

linkedin.com/in/guzzobruno

huggingface.co/bguzzo2k

Soft Skills

- Leadership & Mentoring
- Team & Project Management
- Consulting & Client Engagement
- Agile Methodologies (Scrum)

Technical Skills

Amazon Web Services

- **Services:** Lambda, DynamoDB, Redshift, S3, IAM, VPC
- **Tools:** CDK, CloudFormation, AWS CLI

Google Cloud

- **Compute/AI:** Cloud Run, Functions, Vertex AI, Gemini, BigQuery ML
- **Data/Infra:** BigQuery, SQL, VPC, IAM, Pub/Sub, Cloud Task, Firestore

AI & Machine Learning

- **Tools:** PyTorch, Langchain, Hugging Face, NumPy, Scikit-learn
- **Concepts:** LLM, RAG, NLP, Fine-Tuning, GNNs, Anomaly Detection

DevOps

- **IaC:** Terraform, Fabric
- **CI/CD:** Cloud Build, GitLab CI, GitHub Actions, Artifact Registry
- **Containers:** Docker
- **Others:** Linux, Git, Bash scripting, NginX

Software Development

- **Languages:** Python, Java, C, JavaScript, Typescript, Dart, SQL
- **Frameworks:** Spring Boot, Hibernate, JPA, Flask, Django, SQLAlchemy, Angular, React, Flutter
- **Testing:** Selenium, Junit, PyTest
- **Databases:** PostgreSQL, MySQL, NoSQL
- **Others:** Jira, Sprint Planning, Cloud-Native, Design Patterns, Keycloak

Languages

- **Italian:** Native
- **English:** B2, Cambridge certified